# Yang-Che (Jeff) Tseng

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I build the backend and infrastructure behind AI applications, LLM and computer vision, that take repetitive work off people's hands. Shipped a serverless email pipeline that has run in production for over a year (up to 35k emails/month), built agentic tooling for HPC experiments, and ran 110k+ GPU jobs.

## Education

- **University of British Columbia** — Average 92.8%
  *M.Sc. in Computer Science – Advisor: Kwang Moo Yi* — *Sep 2024 – Nov 2026*
- **National Taiwan University** — GPA 3.92/4.0
  *B.Sc. in Computer Science and Information Engineering* — *2019 – 2023*

## Experience

- **University of British Columbia** — Vancouver, Canada
  *Master of Science, Computer Science* — *Sep 2024 – Nov 2026*
  *Research Assistant, Computer Vision Lab* — *2023 – 2024*
  - **Agentic HPC ops:** built a **Claude Code** harness (48 skills, 6 subagents, safety hooks, post-mortem rules) that takes experiments from code change to PR: ports baselines, submits, monitors and fixes Slurm jobs over SSH, and parses results, so researchers stop babysitting experiments. Parallel sessions opened **281 PRs**; **227 were merged** after review.
  - **GPU fleet:** ran **110k+ Slurm jobs (≈15.8k GPU-hours)** across 3 providers (lab-owned, UBC Sockeye, Compute Canada), peaking at **127 concurrent GPUs**; packaged baselines as **Apptainer** images.
  - **Job scheduling:** sized eval jobs from measured per-frame cost to fit the 3 h limit: **864 jobs** instead of ∼11.6k.
  - **Throughput:** cut per-scene eval for CogNVS from ∼3.5 h to ∼21 min (batched across GPUs) and Lyra from ∼3.75 h to ∼1 h (6 trajectory jobs in parallel).
  - **Research:** showed that single-image NVS benchmarks score misaligned predictions; proposed matching → PnP → co-visibility mask and re-benchmarked **11 baselines** on 4 datasets (1.7k test views).
- **MyProGuide (TravoAI)** — Remote
  *Software Engineer (Part-time)* — *Jul 2024 – Mar 2025*
  - **Serverless email pipeline:** built an **AWS SES → Lambda → S3** pipeline with parsers for 12 OTAs across 4 countries (Klook, KKday, Viator, . . . ) that reads cancellation emails and updates orders via API, so Ops no longer processes cancellations by hand; **in production 1+ year** at up to 35k emails/month.
  - **LLM tooling:** prototyped a **text-to-SQL** agent (LangChain, Gemini) that creates PostgreSQL test accounts.
  - **Flutter app:** built a tour-guide app from scratch through spec-first issues and reviewed PRs (BLoC, OAuth, CI).
- **GoGoX** — Taipei, Taiwan (remote from Vancouver)
  *Software Engineer, Data Scientist* — *Jun 2023 – Jun 2024*
  *Data Scientist Intern* — *Sep 2022 – Jun 2023*
  - **LLM support bot:** prototyped a RAG chatbot (**OpenAI, LangChain, Pinecone**) on **GCP** for routine customer-service questions; cut latency by 2 s with **Redis Stack**.
  - **Driver stopping model:** extracted the **GraphHopper** routing engine from a deprecated Java dispatch system into a **Docker Compose** service with a Python interface; ran counterfactual analysis on driver stop probability.
  - **Labeling automation:** ML model on GPS/order data that generates pickup-time labels that didn't exist before, with **autoencoder** anomaly filtering.
  - **GPS map matching:** **HMM** matching to OpenStreetMap (Python, shapely) to build a travel-time database for ETA.
- **ASUS Intelligent Cloud Services (AICS)** — Taipei, Taiwan
  *Software Engineering Intern* — *Jul 2022 – Sep 2022*
  - Extended Selenium IDE with **Electron / Node.js**, so QA could automate 19 of 23 test cases previously run by hand.

## Publications

- **WACV 2027** – *Aligning Misaligned Benchmarks for Single-View Novel View Synthesis.* **Y.-C. Tseng**, E. Dong, J.-g. Kwak, G. Sharma, S. Kheradmand, S. Garbin, H. Isack, K. M. Yi.
- **NeurIPS 2024** – *3D Gaussian Splatting as Markov Chain Monte Carlo.* S. Kheradmand, D. Rebain, G. Sharma, W. Sun, **Y.-C. Tseng**, H. Isack, A. Kar, A. Tagliasacchi, K. M. Yi.
- **ECCV 2024** – *PointNeRF++: A Multi-Scale, Point-Based Neural Radiance Field.* W. Sun, E. Trulls, **Y.-C. Tseng**, S. Sambandam, G. Sharma, A. Tagliasacchi, K. M. Yi.
- **ICRA 2023** – *Orbeez-SLAM: A Real-time Monocular Visual SLAM with ORB Features and NeRF-realized Mapping.* C.-M. Chung, **Y.-C. Tseng**, Y.-C. Hsu, X.-Q. Shi, Y.-H. Hua, J.-F. Yeh, W.-C. Chen, Y.-T. Chen, W. H. Hsu.

## Teaching

- **UBC 2025** – *CPSC 425: Computer Vision.* Teaching Assistant.
- **UBC 2024** – *CPSC 259: Data Structures and Algorithms for Electrical Engineers.* Teaching Assistant.
- **NTU 2022** – *Deep Learning for Computer Vision.* Teaching Assistant.
- **NTU 2022** – *Network Administration and System Administration.* Teaching Assistant.
- **NTU 2021** – *Data Structures and Algorithms.* Teaching Assistant.

## Skills

- **Backend & Infra:** AWS, Docker, Slurm, Apptainer, GitHub Actions, GCP, Cloudflare Workers
- **AI:** PyTorch, LLM apps and agents, 3D vision
- **Languages:** Python, C/C++, CUDA, TypeScript/Node.js, Dart/Flutter
- **Spoken:** Mandarin (native), English (fluent)
- **Open source:** added per-user filtering to **nvtop**, a GPU monitoring tool (C).